

Infosys
Springboard

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AirFly Insights: Data Visualization and Analysis of Airline Operations Report

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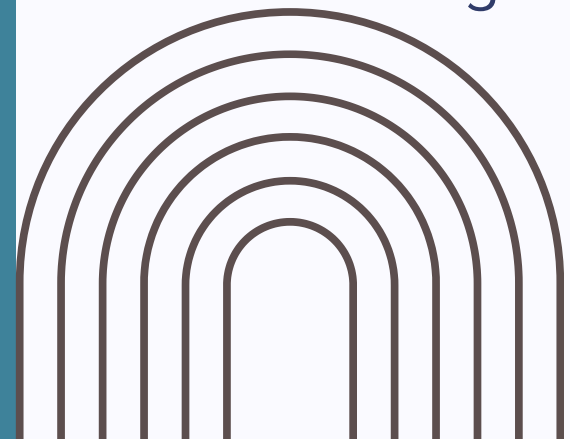
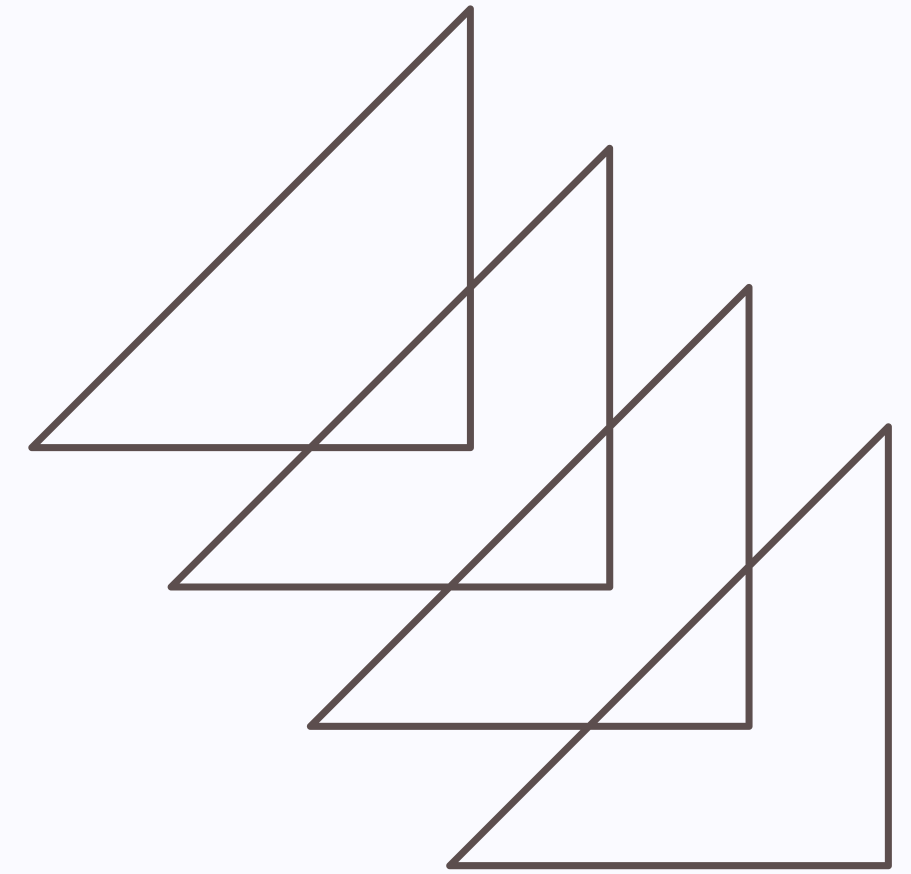
Agenda

- **Project Overview:** Objectives and scope.
- **Data & Methodology:** Dataset summary and analytical approach.
- **Workflow & EDA:** Step-by-step process and initial data patterns.
- **Flight Performance:** Trends by airline, route, and time.
- **Delay & Cancellation Analysis:** Primary causes and seasonal impacts.
- **Technical Stack:** Libraries and tools utilized.
- **Recommendations & Conclusion:** Key takeaways and future steps.

Introduction

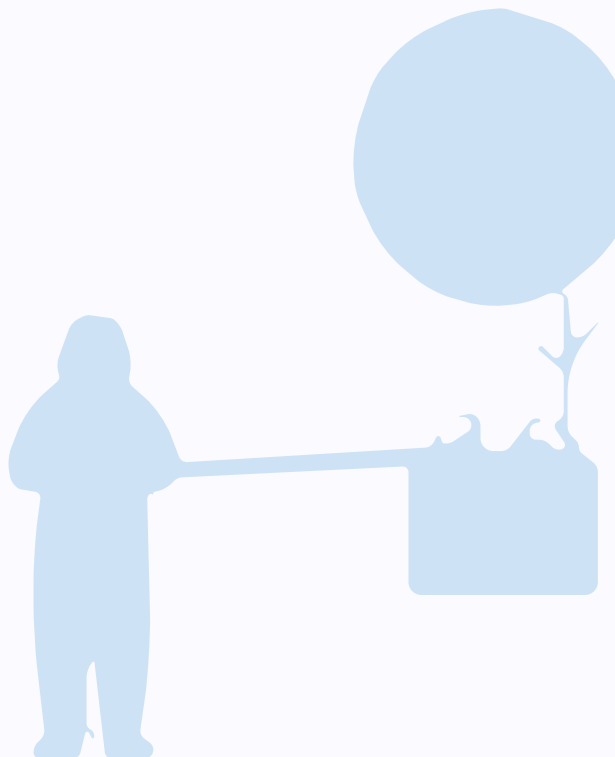
Overview of objectives and problem statement:

- Analyze large-scale airline flight data to identify operational trends
- Study flight delay patterns across airlines and airports
- Examine common reasons for flight cancellations
- Use data visualization techniques to represent insights clearly
- Evaluate airline-level and airport-level performance
- Support data-driven and actionable insights through visual analysis





Expected Outcomes

- Understand and preprocess aviation datasets for analysis
 - Explore trends in flight schedules, delays, cancellations, and routes
 - Visualize key metrics using bar charts, time series, heatmaps, maps, and comparisons
 - Provide insights for stakeholders including airline operators and analysts
 - Summarize findings through a final visual report and presentation
- 

Project Workflow & Objectives



Understand & Preprocess Aviation Datasets for Analysis



Explore Trends in Flight Schedules, Delays, Cancellations, & Routes



Visualize Key Metrics Using Bar Charts, Time Series, Heatmaps, Maps, & Comparisons



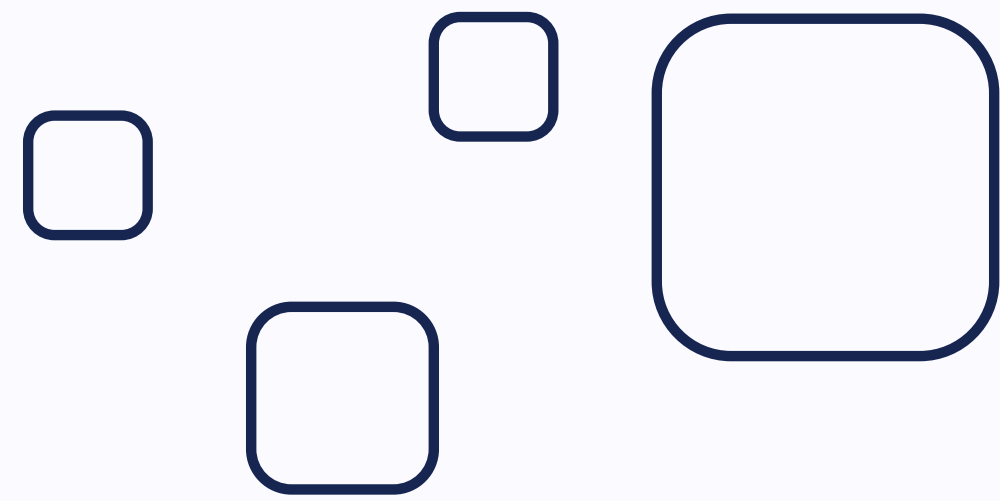
Provide Insights for Stakeholders Including Airline Operators & Analysts



Summarize Findings through Final Visual Report & Presentation

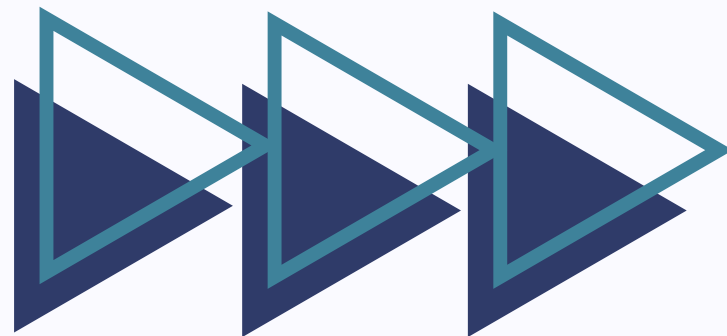
Expected Outcomes:

- Actionable Insights for Operational Improvement
 - Comprehensive Visual Report for Stakeholders
- 



Dataset Overview

Insight into the data source, size,
and key variables analyzed



Source: Kaggle

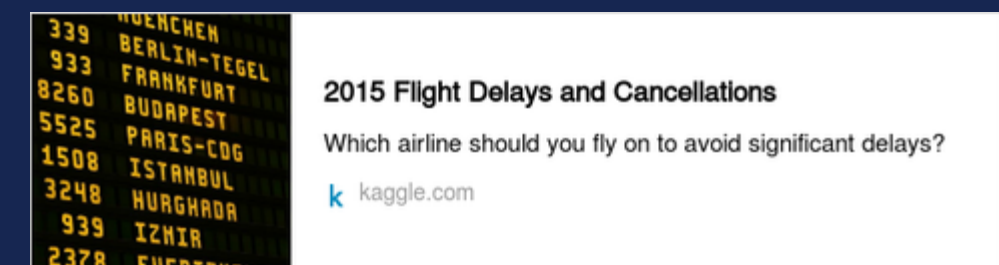


Datasets and Size

1. Flights (5819079 rows, 31 cols)
2. Airports (322 rows, 7 cols)
3. Airlines (14 rows, 2 cols)



Link:

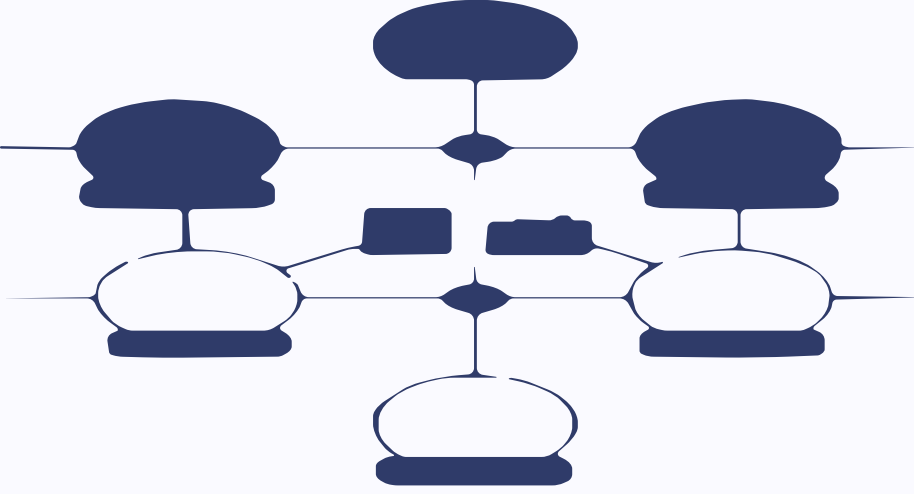


Methodology Overview

Key steps taken in cleaning data and analytical approach utilized

- Data Acquisition and Understanding: Collected and reviewed flight datasets.
- Data Cleaning and Feature Engineering: Cleaned data and created key features.
- Univariate and Bivariate Analysis: Analyzed metrics and relationships.
- Delay Cause Analysis: Identified main delay reasons.
- Cancellation and Seasonal Trends: Studied seasonal cancellation patterns.
- Airport and Route-Level Exploration: Compared airports and routes.
- Final Dashboard or Report: Visualized insights clearly.
- Documentation and Presentation: Documented and presented results.

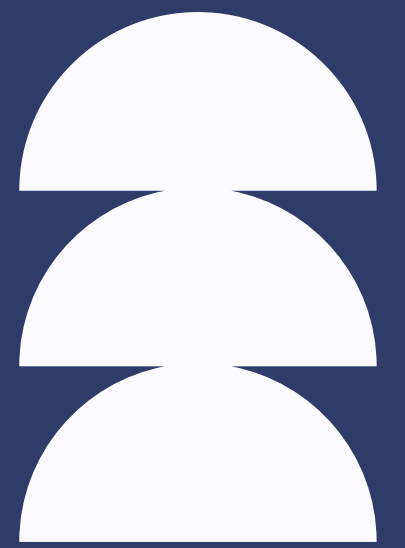




Project Workflow



Exploratory Analysis

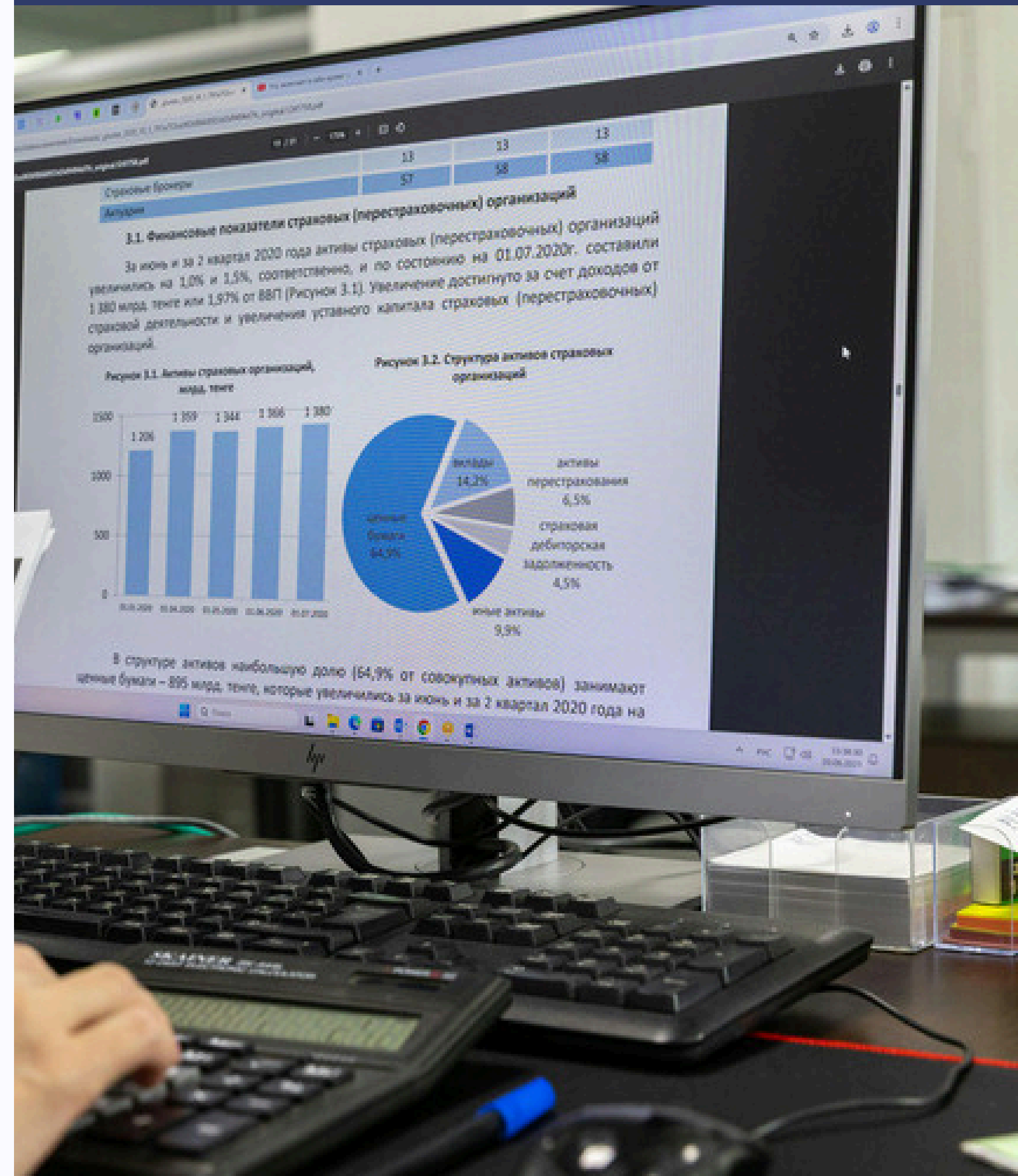


Key Patterns and Insights from Data Exploration Techniques

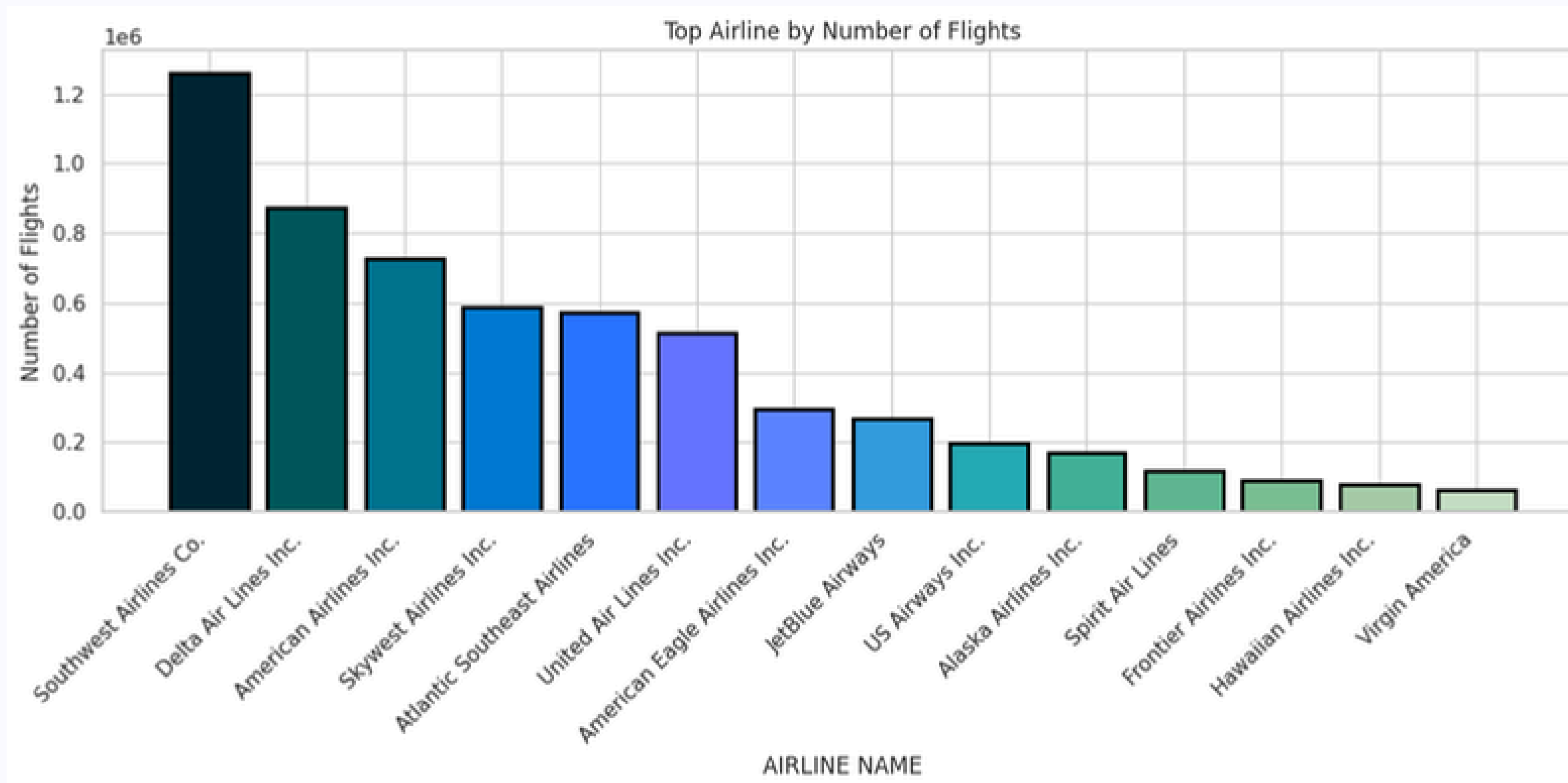
- Defined project goals, and overall workflow
- Loaded CSV datasets using pandas
- Explored schema, data types, size, and null values
- Performed data sampling and memory optimization
- Handled null values in delay and cancellation columns
- Created derived features such as month, day, hour, and route
- Formatted and standardized datetime columns
- Performed Feature engineering, Ex: Date, Route etc

Visualization Highlights

Next Milestones focused on visual analysis to uncover delay, route, and cancellation patterns across airlines, airports, and time periods. The analysis highlighted peak delay times, delay-prone carriers, congested routes, and strong seasonal effects on cancellations, supported through multiple data visualizations.



Uni & Bivariate Visuals & thier Findings: Top Airlines

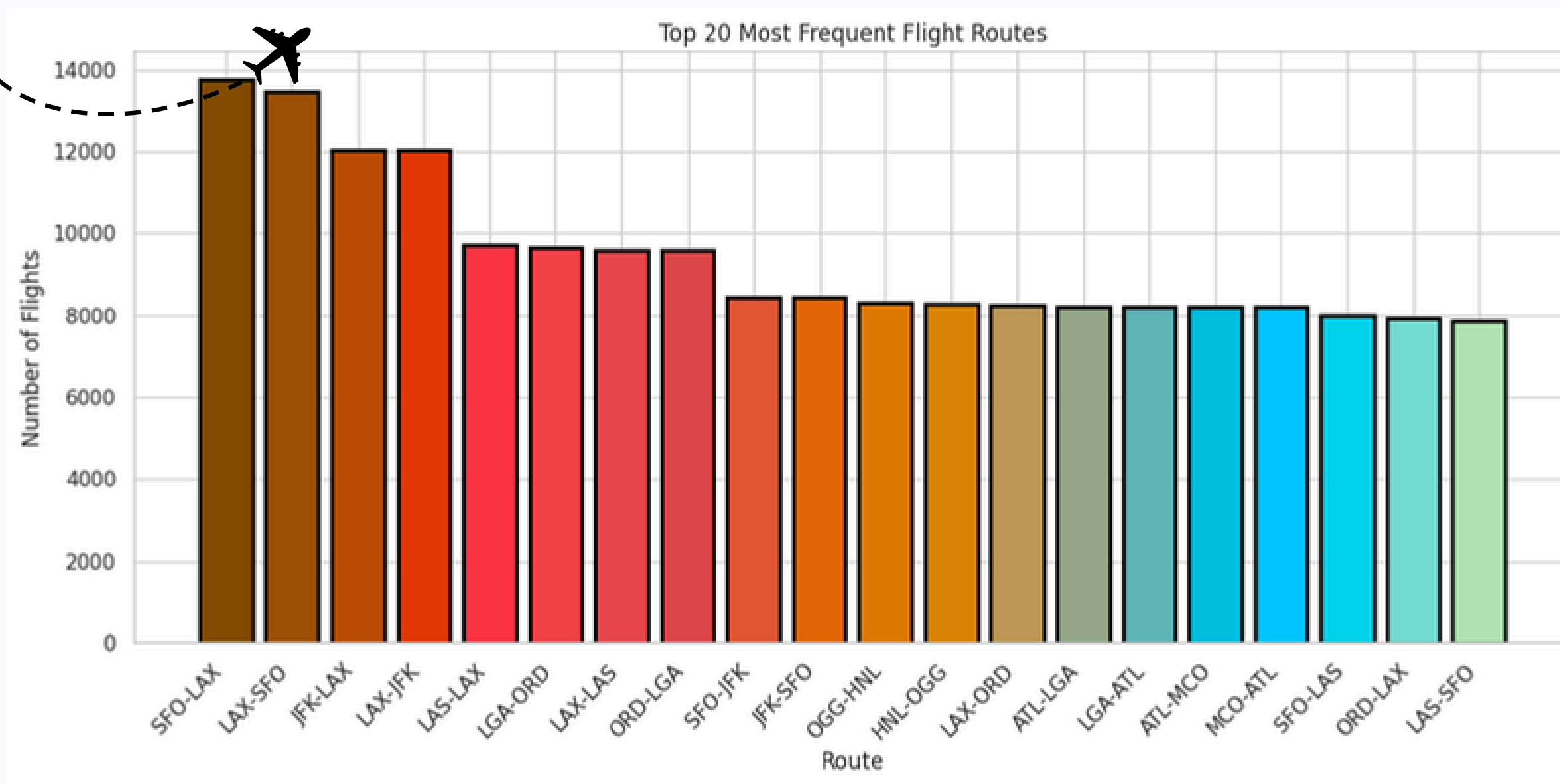


Business Context

Southwest (WN) is the busiest airline, followed by Delta and American, with the top five carriers handling over 65% of all flights. Smaller airlines contribute minimally, so operational issues with major carriers have the greatest system-wide impact.



Top Routes



Insights and Suggestions:

- High-traffic routes like LAX–SFO and JFK–LAX dominate operations, indicating strong demand on major hub-to-hub corridors and key revenue-generating routes.
- Optimizing scheduling, capacity, and delay management on these routes can significantly improve efficiency, punctuality, and overall profitability.



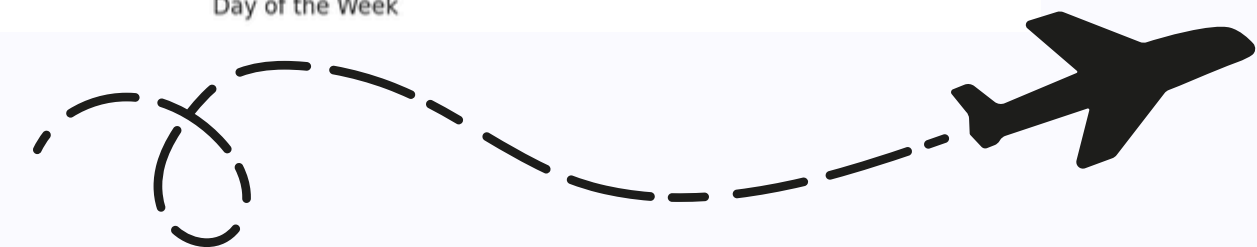
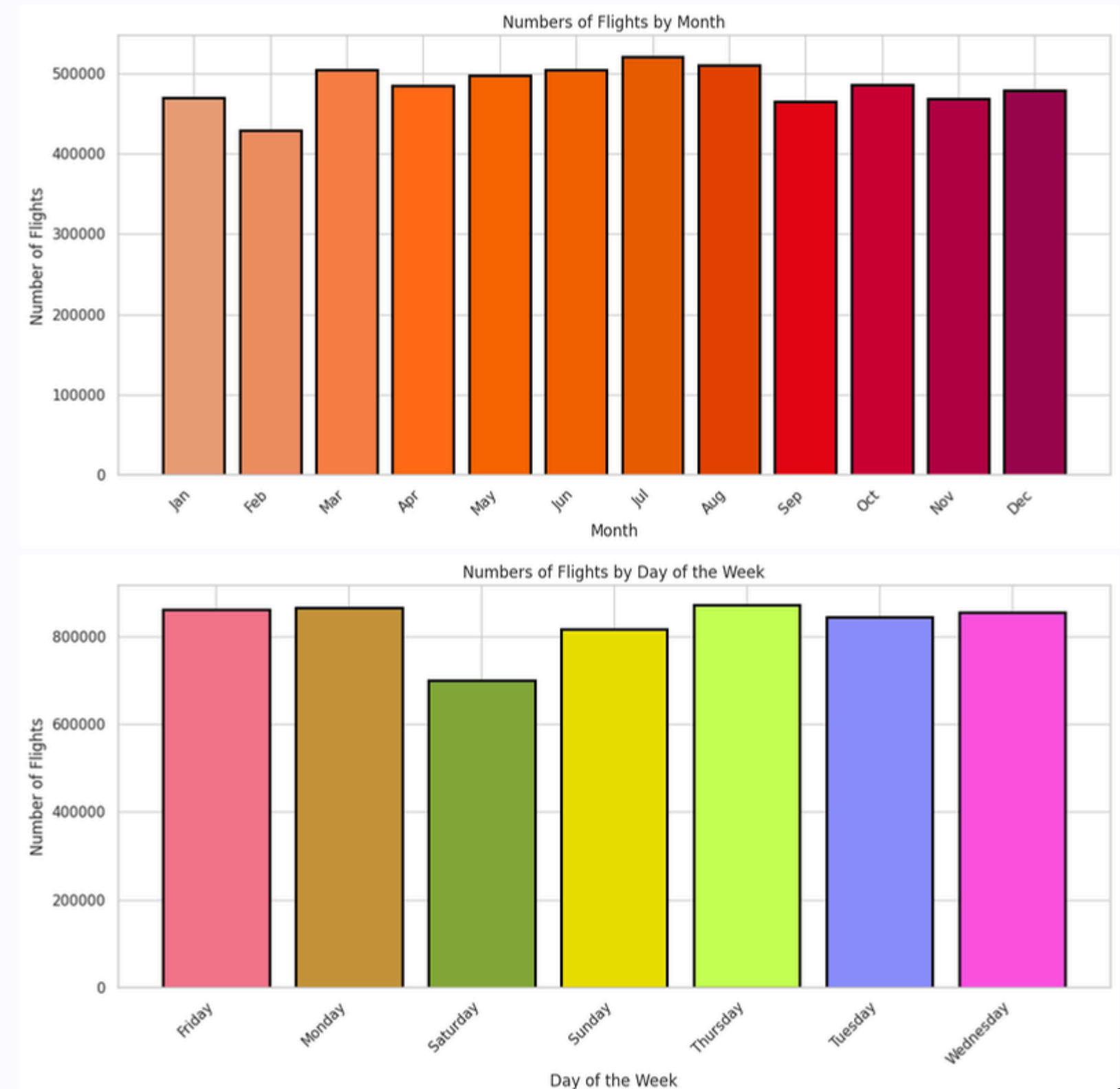
Flight Distribution by Month & Day

Flights by Month:

- Flights peak in mid-year, especially July and August.
- Early and late months show relatively lower traffic.
- Plan resources and pricing for higher summer demand.

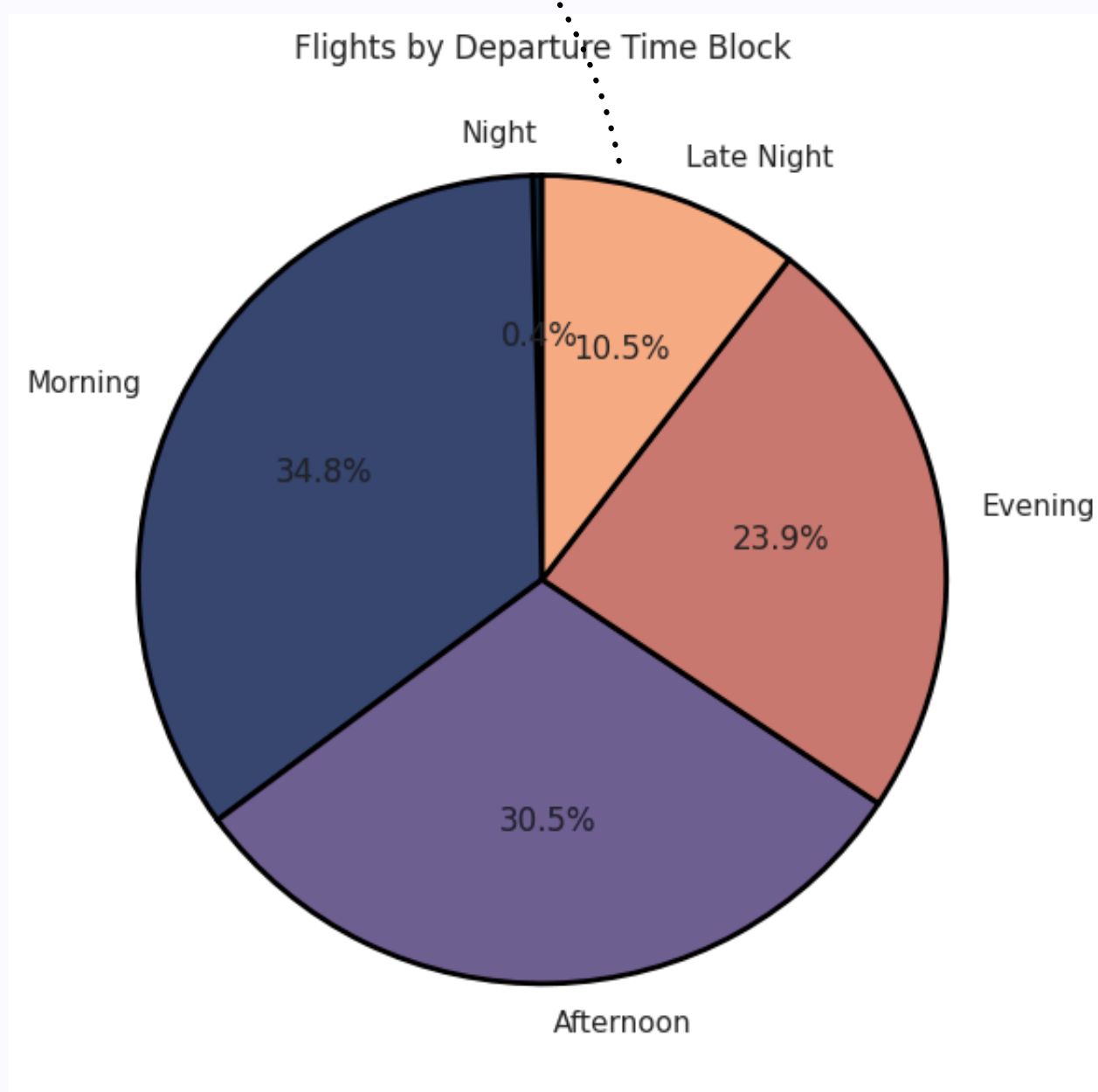
Flights by Day:

- Weekdays have higher flight volumes.
- Saturday is the least busy day.
- Optimize schedules and maintenance on weekends.





Flights by Departure Time



Findings

- Most flights operate in the morning and afternoon, together accounting for nearly two thirds of total traffic.
- Evening flights have moderate demand, while late night and night flights are very low.

Suggestions to Improve Business

- Increase capacity, pricing optimization, and premium services during morning and afternoon peak hours.
- Use discounts or special offers to boost demand for evening and late-night flights.
- Schedule maintenance or low priority routes during night hours to reduce operating costs.

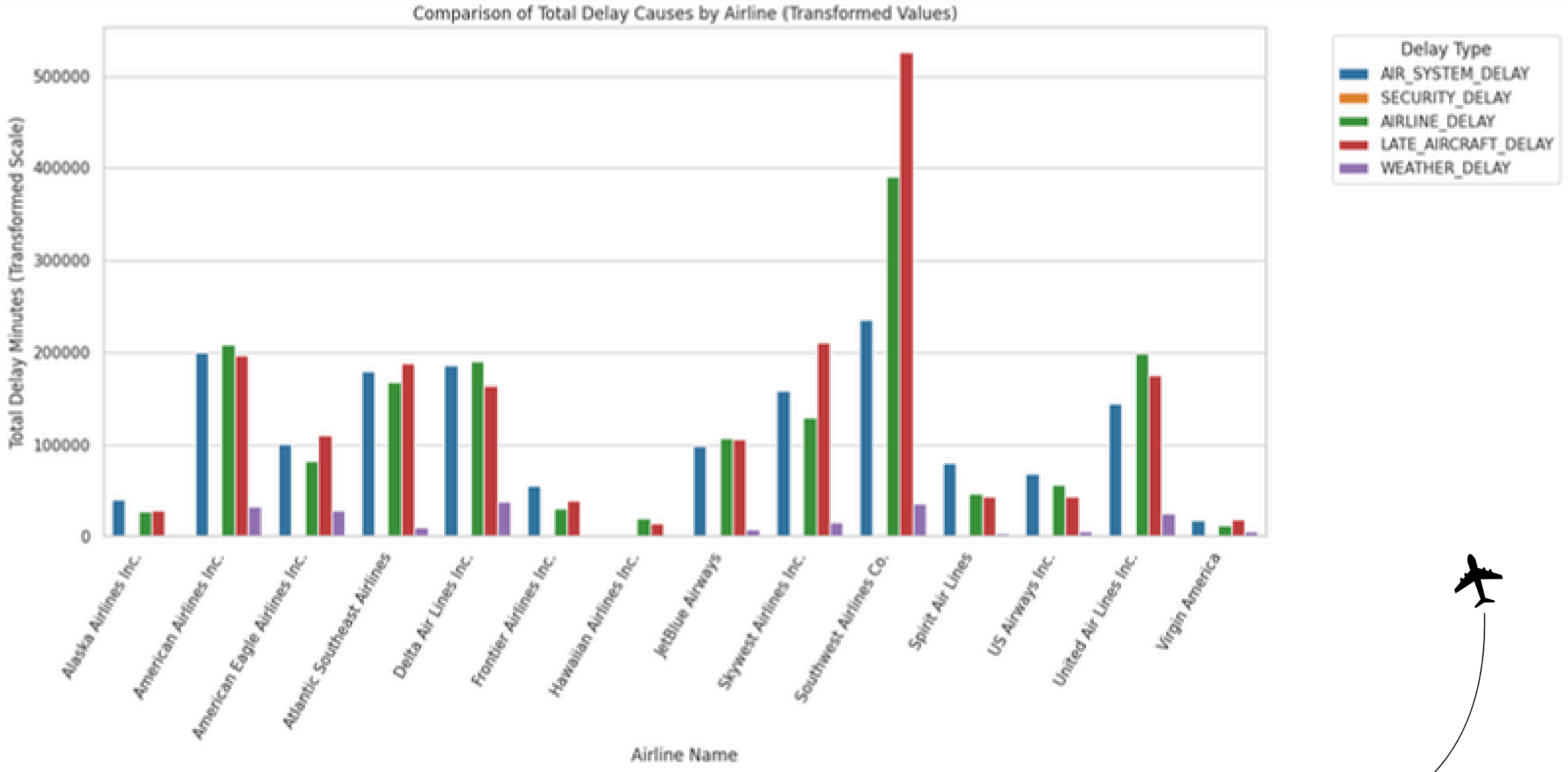
Comparision of Delay Causes by Airline

Findings

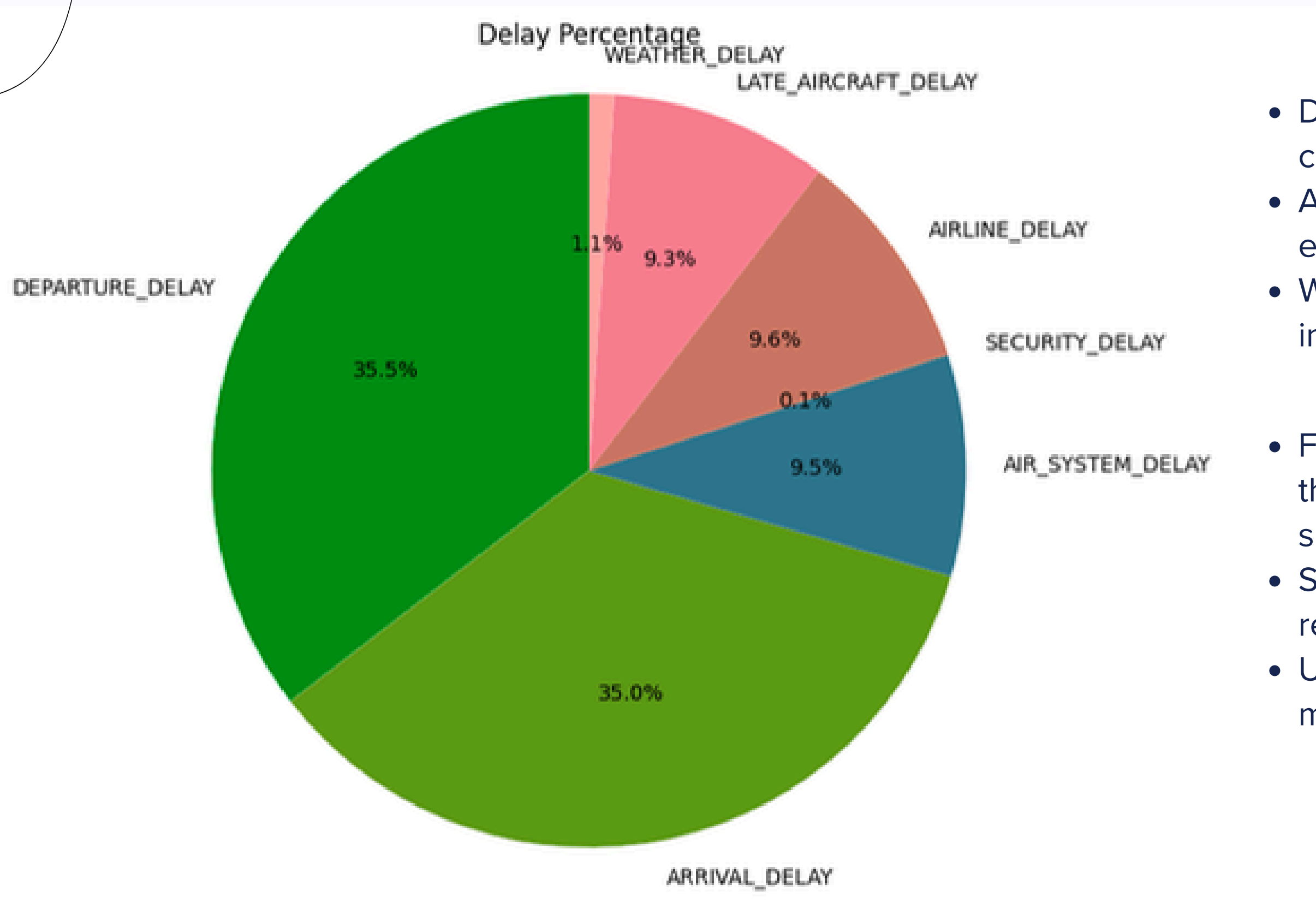
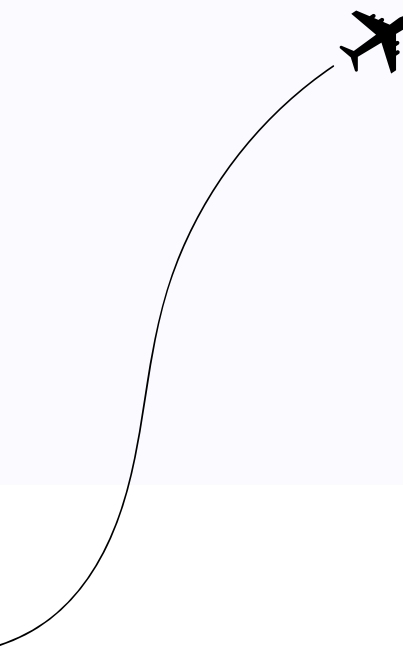
- Late aircraft and airline-related delays are the largest contributors across most airlines.
- Major carriers show significantly higher total delays due to higher flight volumes.
- Weather and security delays are comparatively low and less impactful.

Suggestions to Improve

- Improve aircraft turnaround time and crew scheduling to reduce late aircraft delays.
- Use predictive maintenance and real-time operations tracking to minimize airline delays.
- Strengthen contingency planning during peak periods to handle high traffic efficiently.



Delay % by Category



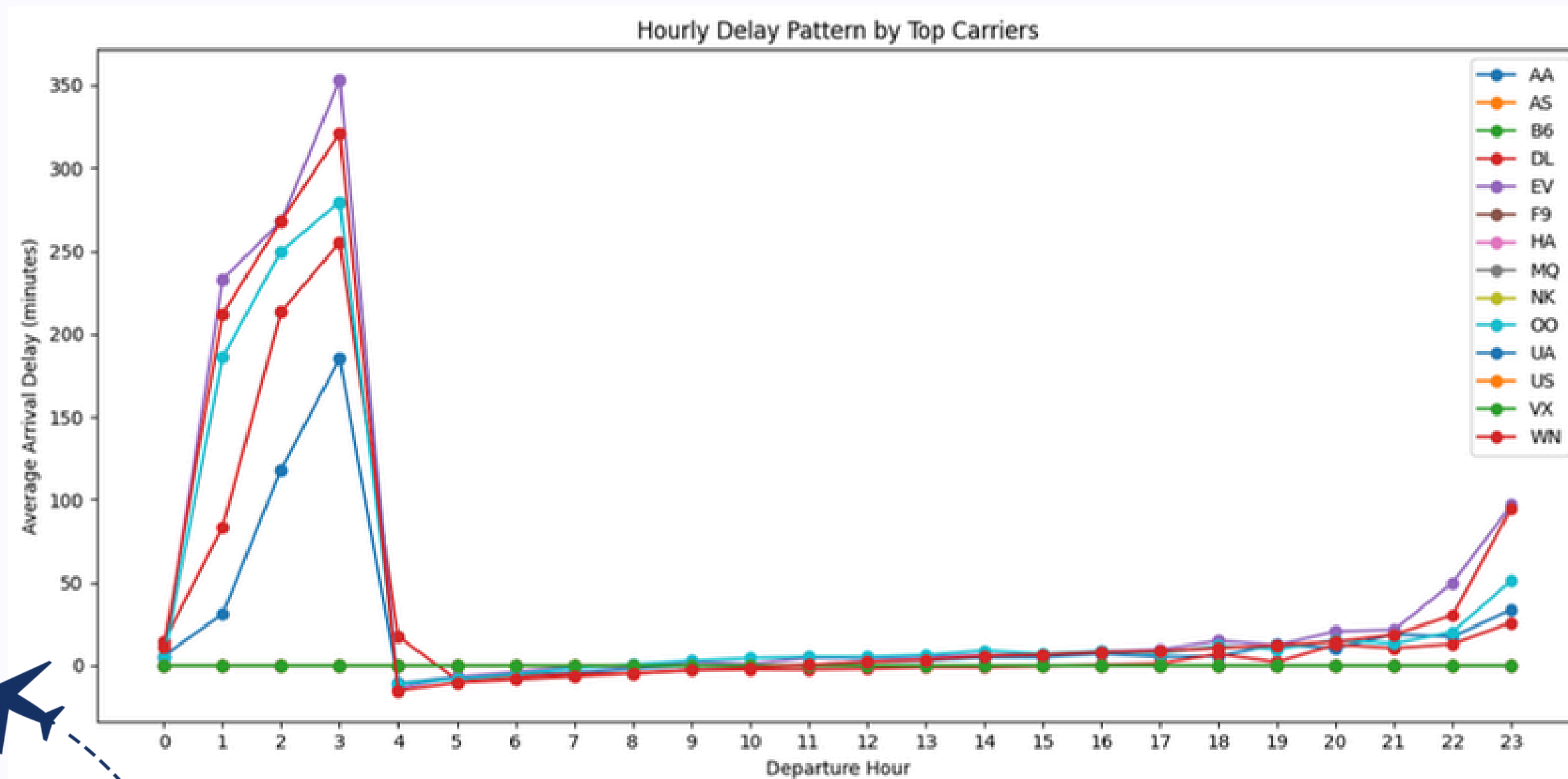
Findings

- Departure and arrival delays together contribute to about 70% of total delays.
- Airline, air system, and late aircraft delays each contribute a moderate share.
- Weather and security delays have minimal impact overall.

Suggestions

- Focus on improving on-time departures through better ground operations and scheduling.
- Strengthen turnaround coordination to reduce arrival-to-departure delay spillover.
- Use data-driven planning to proactively manage airline and air system delays.

Visualizing delays by time of day and Airline



Findings

- Delays peak in early morning (1–4 AM) and rise again late at night.
- Midday hours show relatively stable and lower delays across carriers.

Suggestions

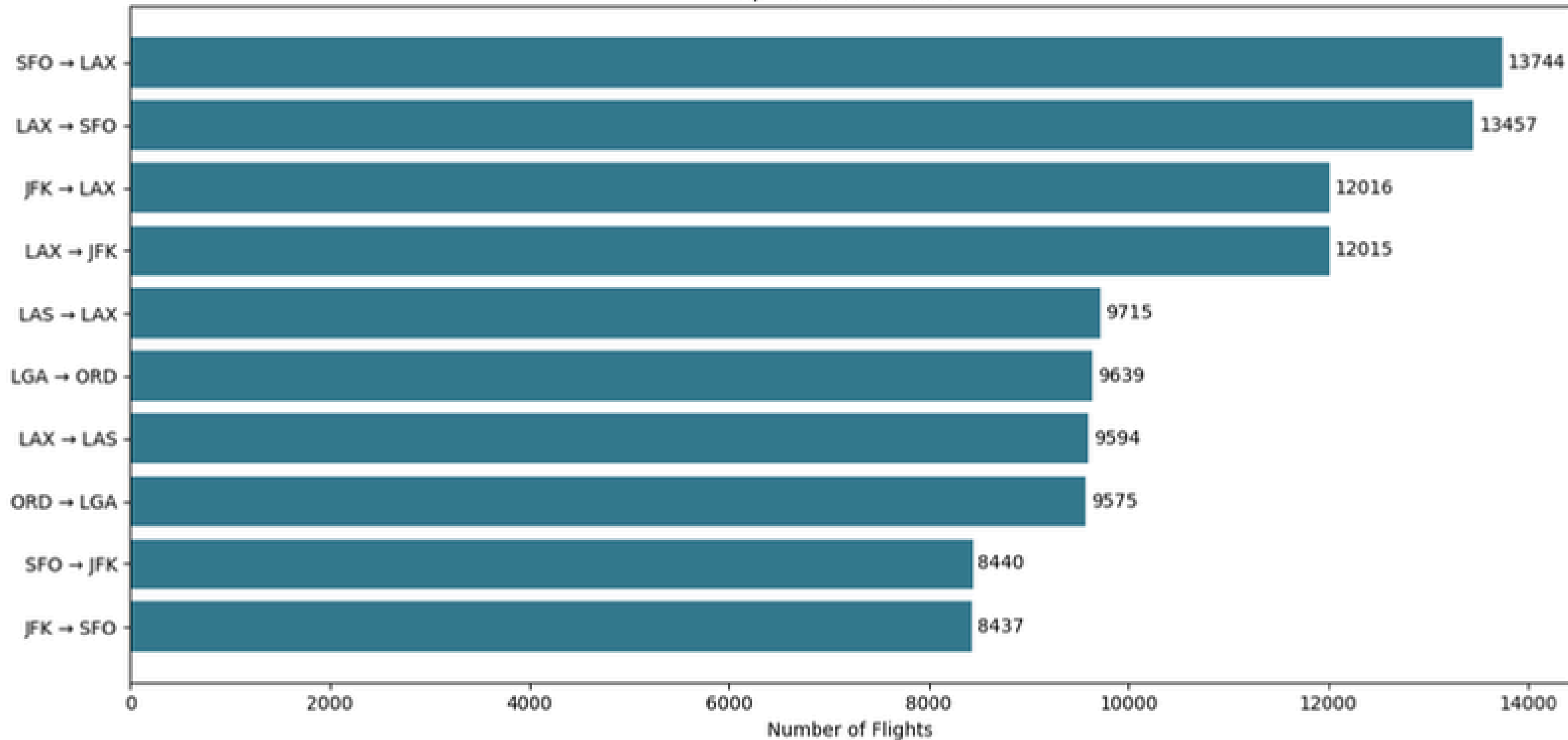
- Strengthen early-morning operations to prevent delay buildup.
- Schedule buffer time and resources for late-night flights to reduce spillover delays.



Top Busiest Routes



Top 10 Busiest Routes



Insights and Suggestions:

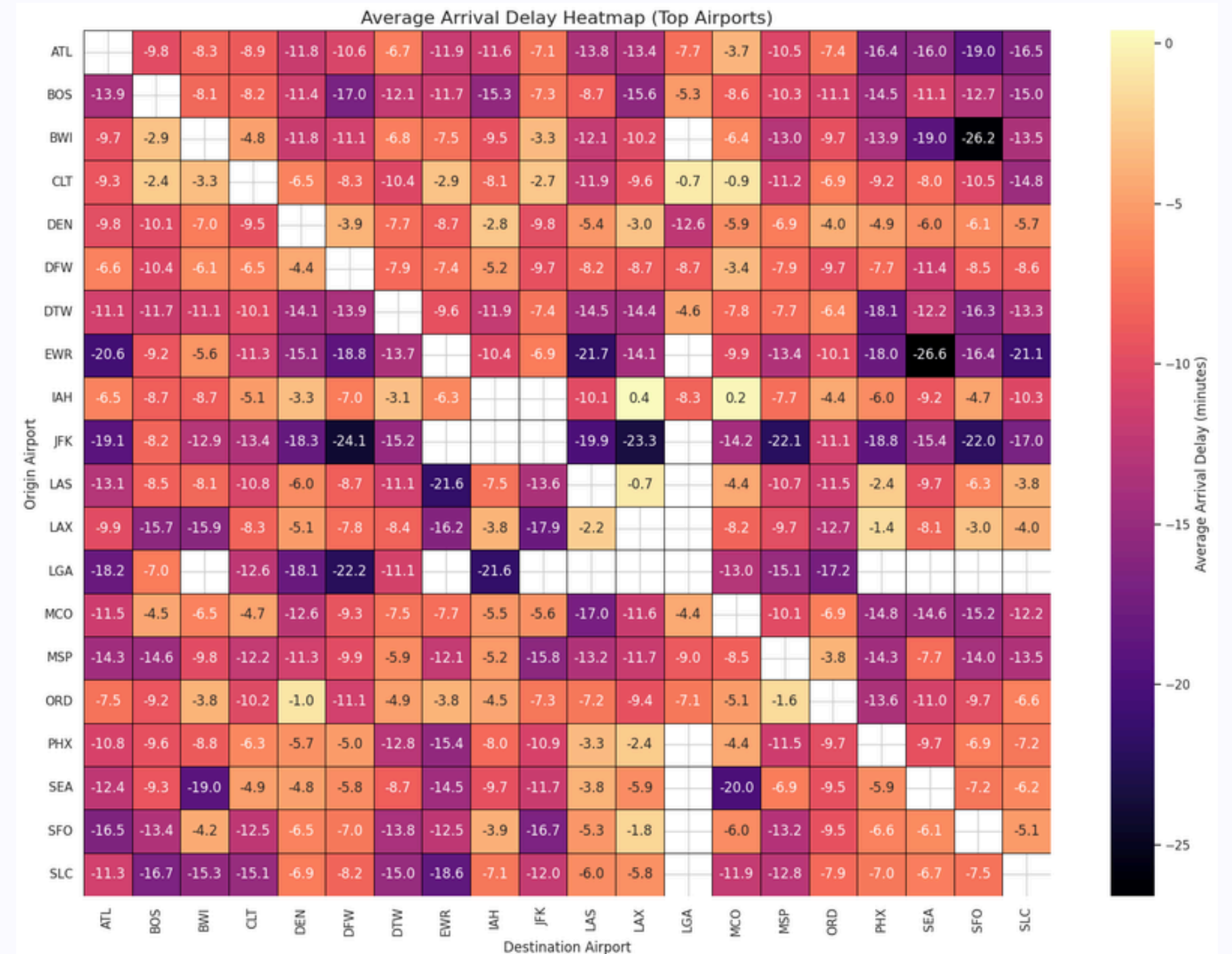
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- Optimizing scheduling, capacity, and delay management on these routes can significantly improve efficiency, punctuality, and overall profitability.



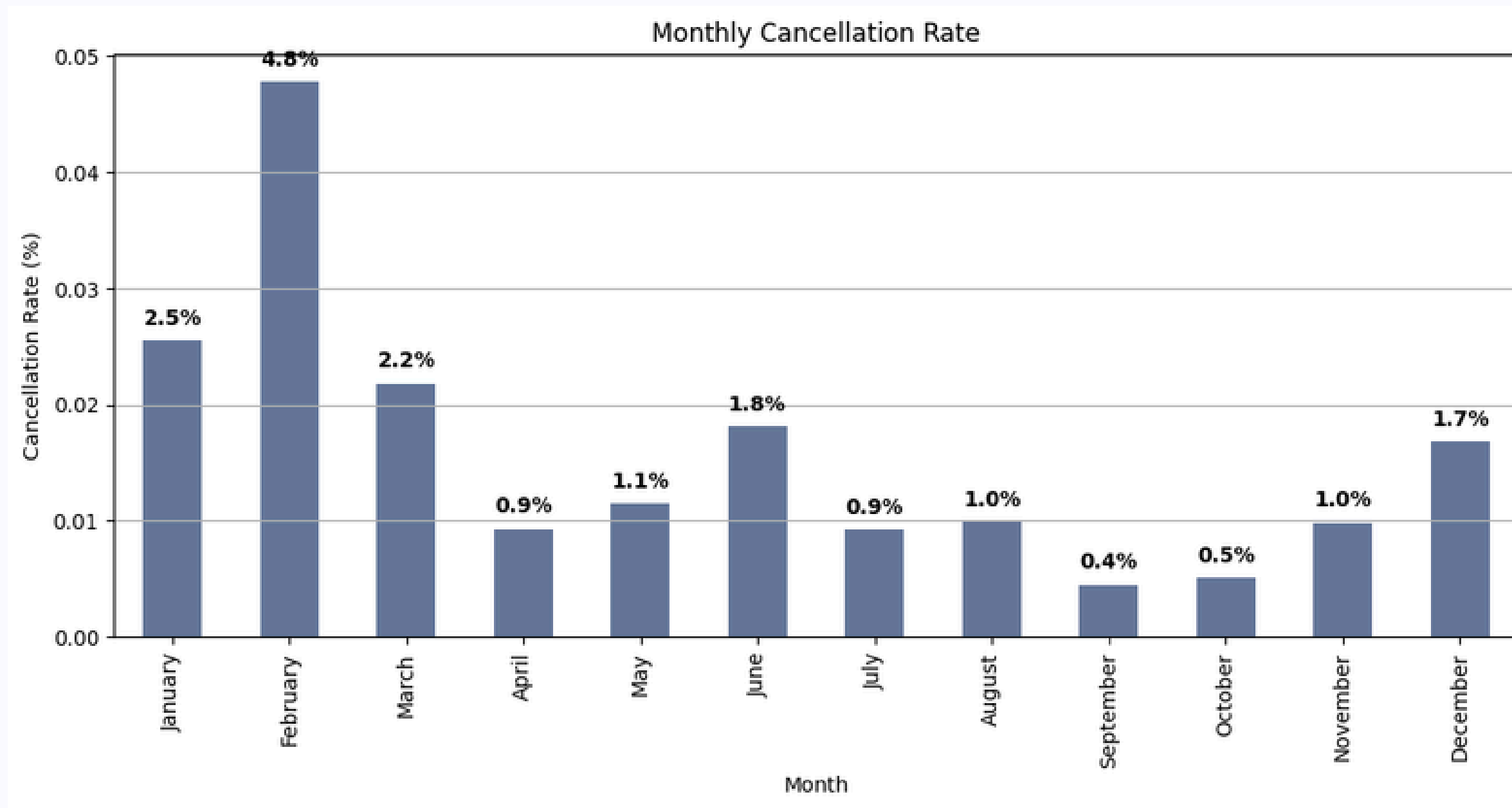
Average Arrival Delay Heatmap

Findings

- **Identify Problematic Routes:** Brighter colored cells indicate routes with higher average arrival delays. These routes may require operational review, resource allocation, or adjusted scheduling to improve on-time performance and customer satisfaction.
- **Leverage Efficient Routes:** Darker colored cells signify routes with lower or even negative (early) average delays. These routes demonstrate strong operational efficiency, which can be leveraged for reliable service and potentially optimized schedules or marketing.
- **Strategic Planning:** Understanding delay patterns helps in strategic planning, such as optimizing hub connections, anticipating resource needs, and informing customer communication regarding potential travel disruptions or reliable routes.



Monthly Cancellation Trends & Rate



Findings

- Cancellation rates are highest in February and January.
- Rates drop significantly during mid-year, with the lowest in September and October.

Suggestions

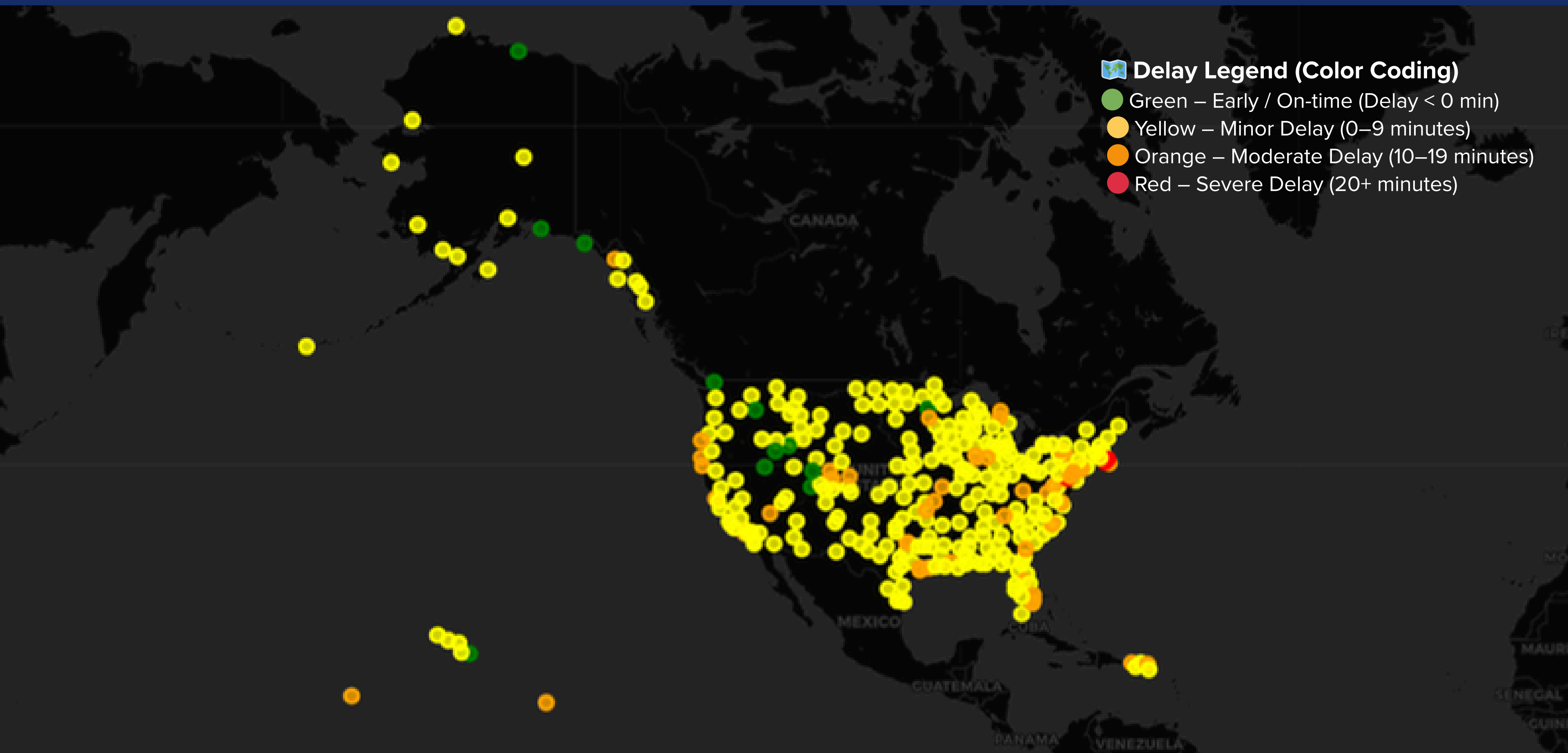
- Strengthen winter weather contingency planning to reduce early-year cancellations.
- Maintain current operations during low-cancellation months and apply best practices year-round.

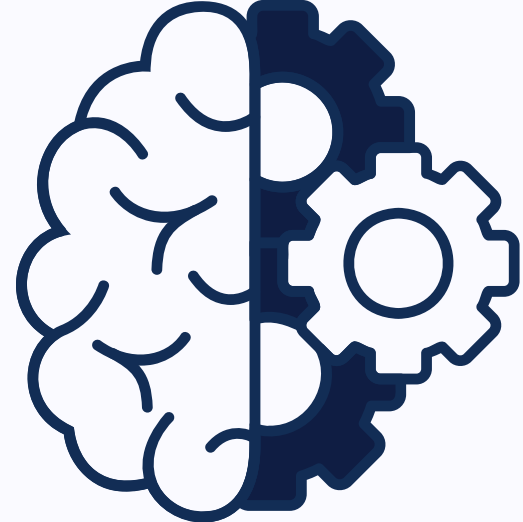




Map Showing Average Delay

-  **Delay Legend (Color Coding)**
-  Green – Early / On-time (Delay < 0 min)
 -  Yellow – Minor Delay (0–9 minutes)
 -  Orange – Moderate Delay (10–19 minutes)
 -  Red – Severe Delay (20+ minutes)





Cancellation Reasons & Percentage



Key Findings

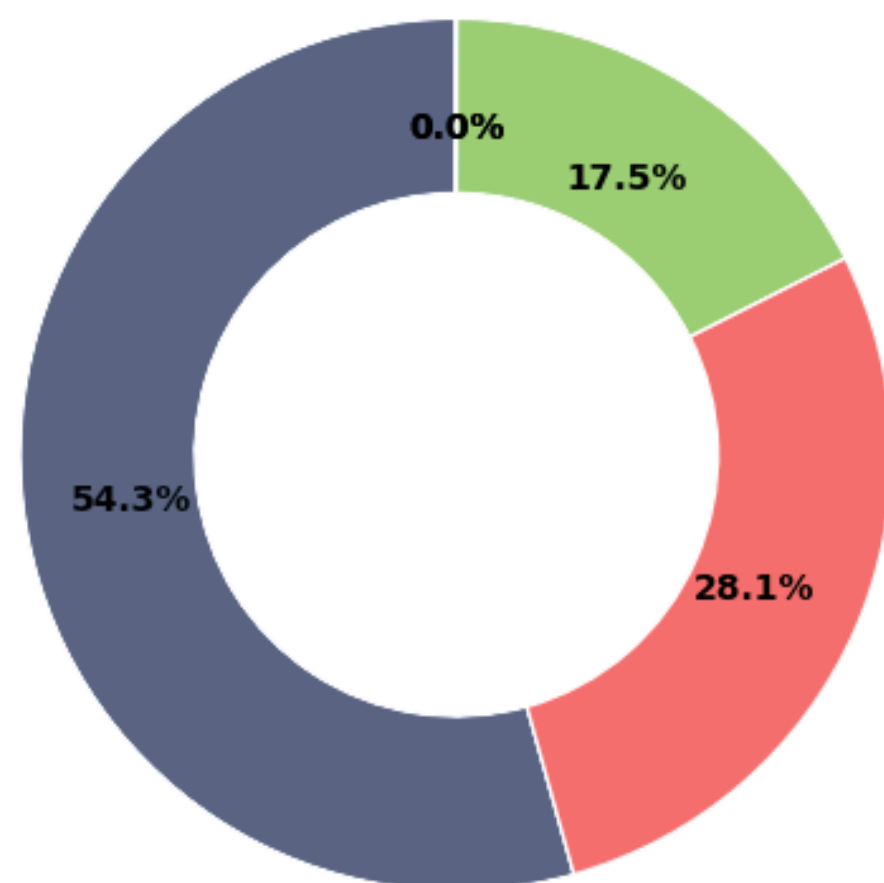
- Weather causes the majority of cancellations (54.3%), making it the biggest risk factor.
- Carrier-related issues (28.1%) are the largest controllable cause.
- NAS delays (17.5%) indicate congestion and airspace dependency.



Business Improvement Suggestions

- Use predictive weather analytics to reschedule flights early and reduce disruptions.
- Improve crew & maintenance planning to minimize carrier-related cancellations.
- Optimize flight scheduling and routing to avoid congested airspaces.

Cancellation Reasons — Donut Chart



Reasons		
Weather	48851	(54.3%)
Carrier	25262	(28.1%)
NAS	15749	(17.5%)
Security	22	(0.0%)
Not Cancelled	0	(0.0%)



Winter Months & Holiday Analysis



Findings

- Winter has the highest cancellation rate (2.9%)
- Spring and Summer show lower and stable cancellations



Suggestions

- Improve winter weather planning
- Add extra buffer flights and crew in winter



Findings

- New Year rush has the highest cancellations (2.55%)
- Christmas and Thanksgiving show moderate impact

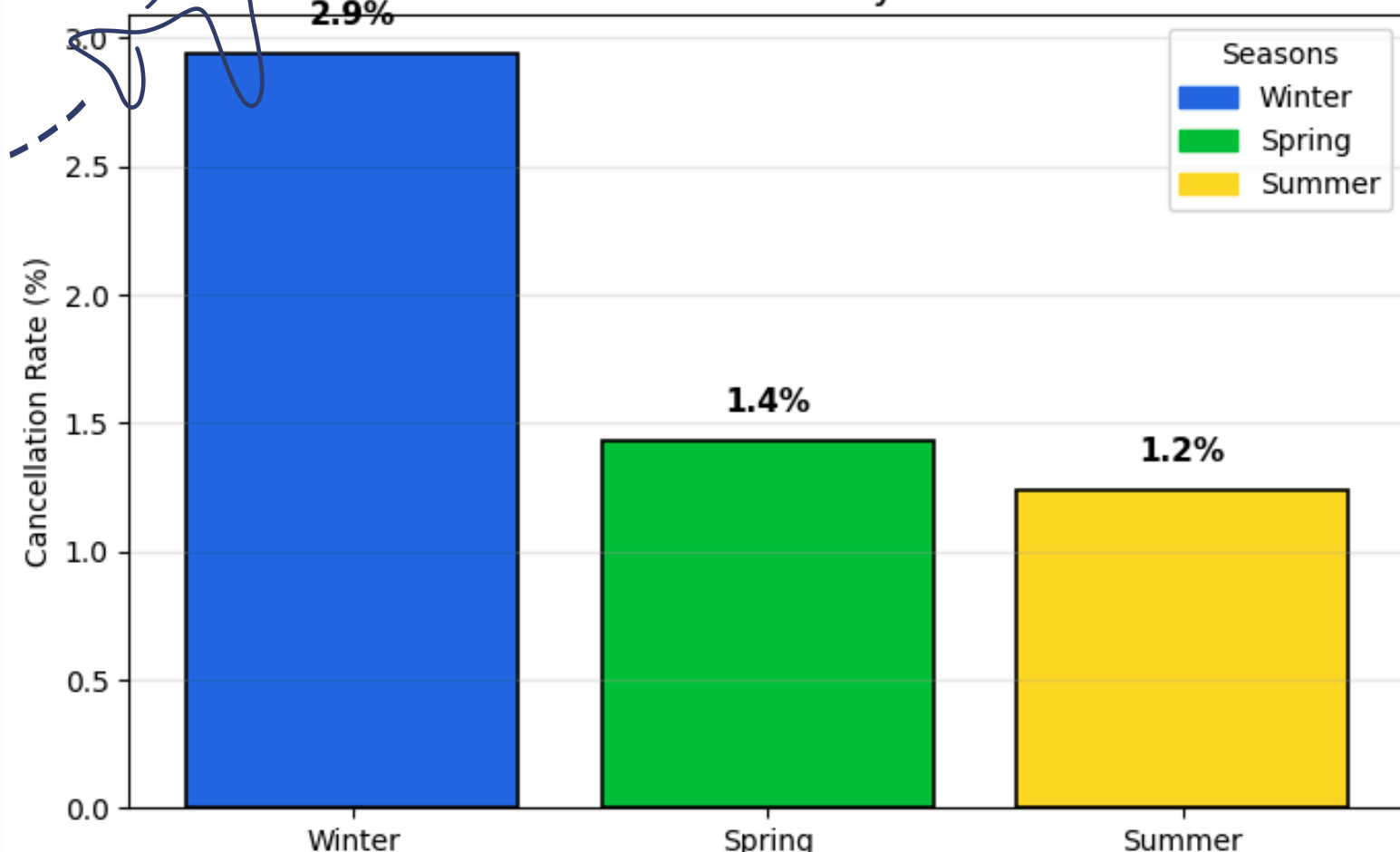


Suggestions

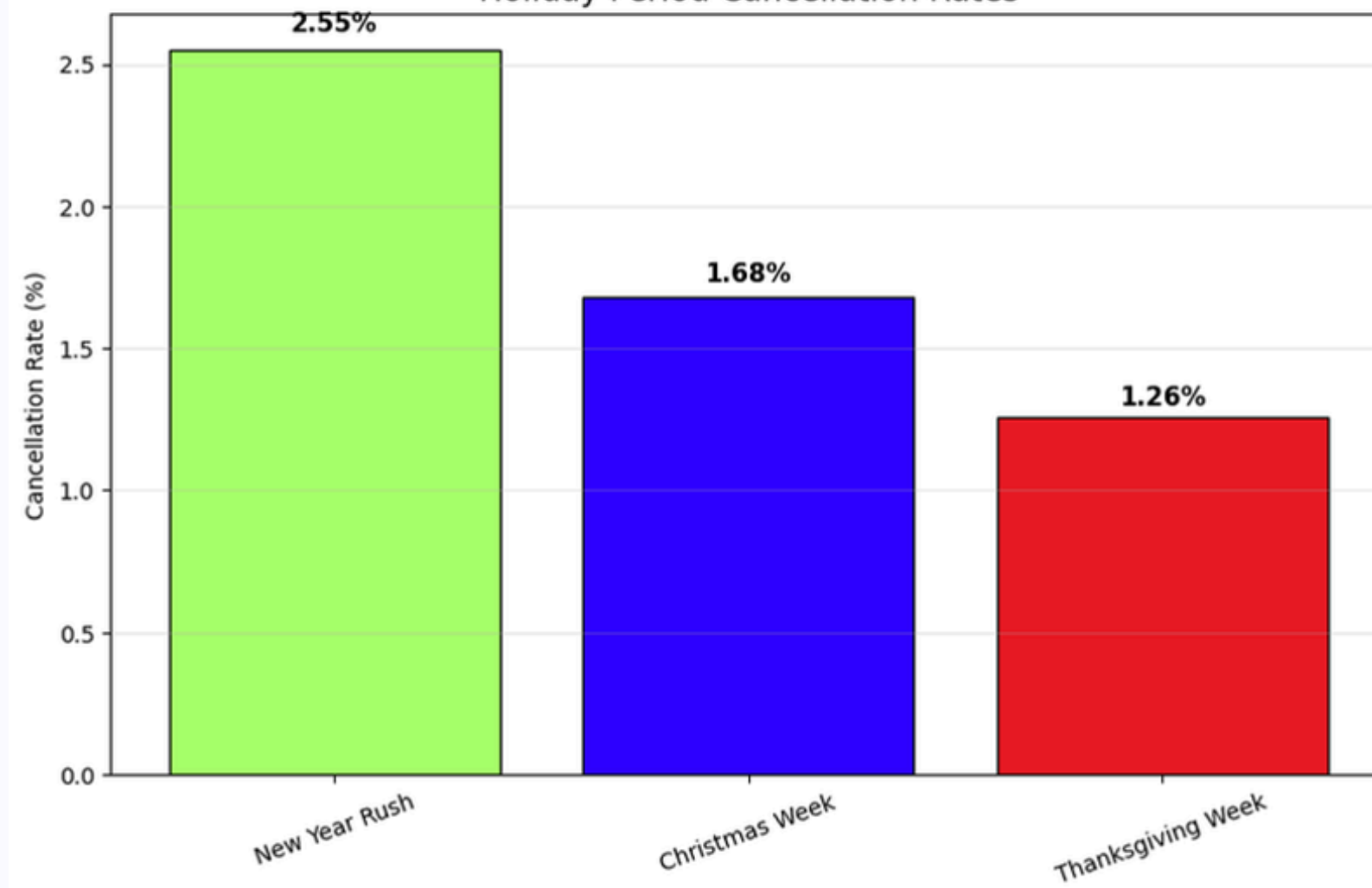
- Increase staffing and backup flights during holidays
- Use advance demand forecasting to reduce disruptions



Cancellation Rate by Season



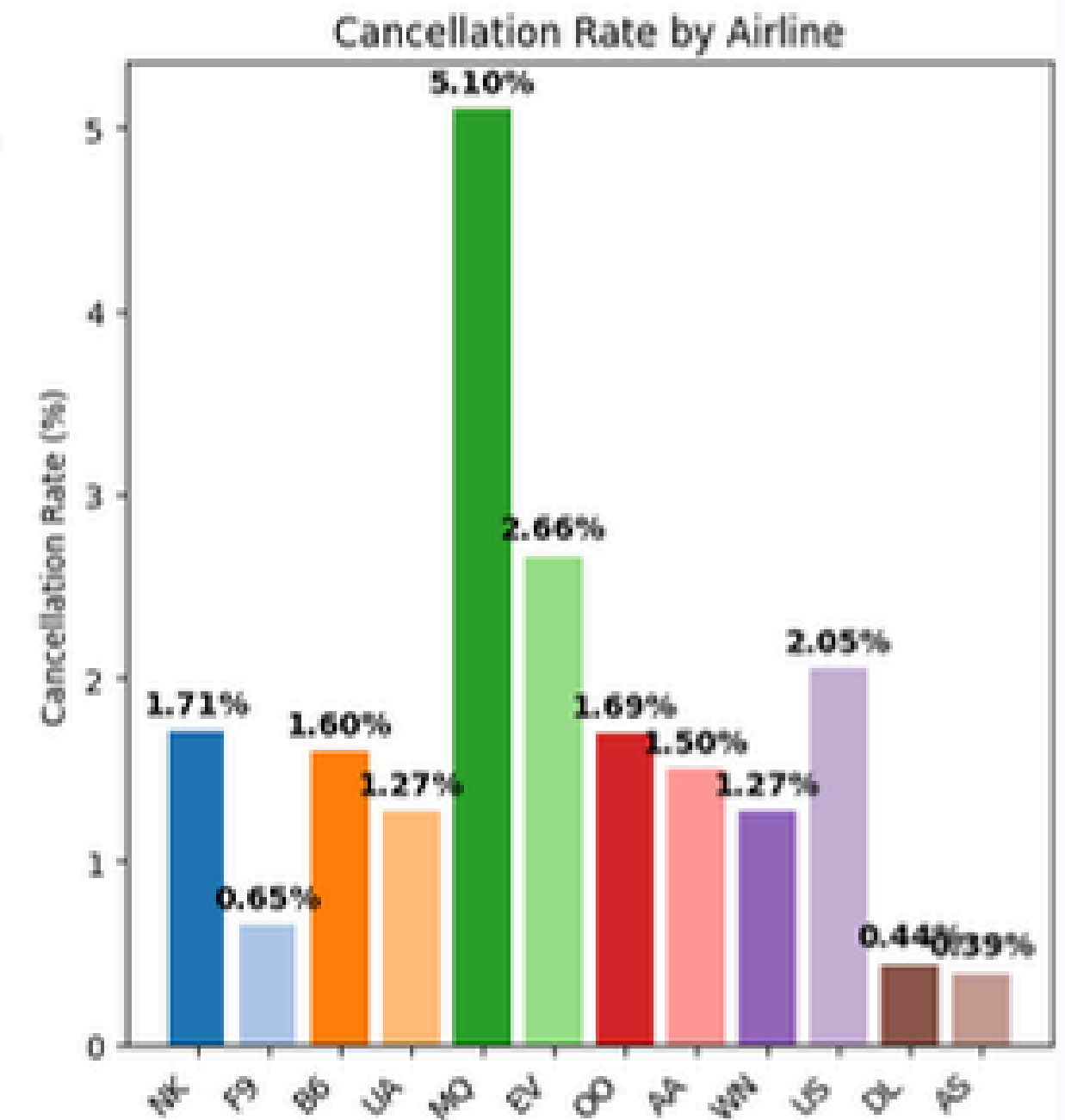
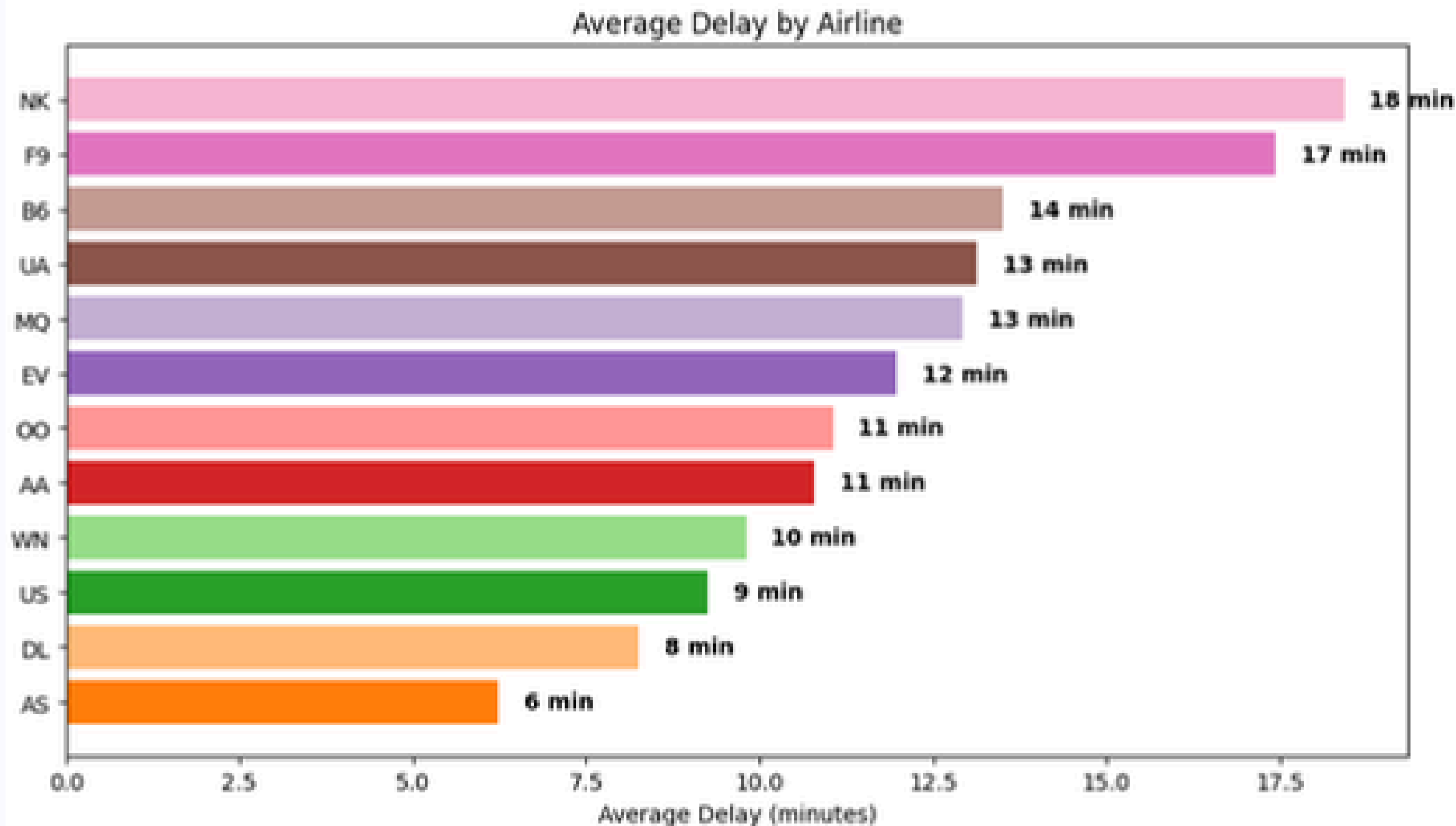
Holiday Period Cancellation Rates





Average Delay & Cancellation Rate by Airline

Airline Performance: Delay & Cancellation



Weather Trend Delay My Month

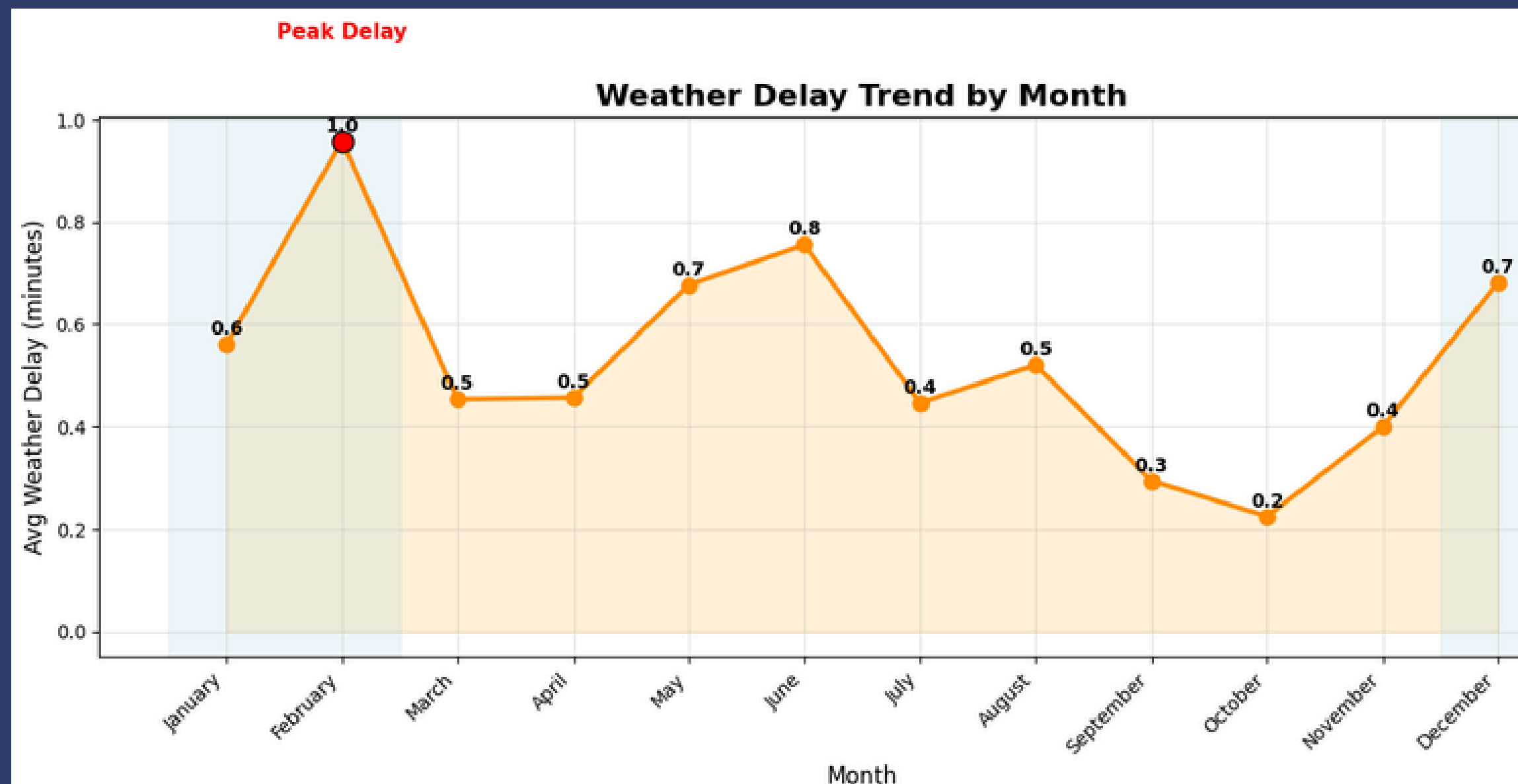


Findings

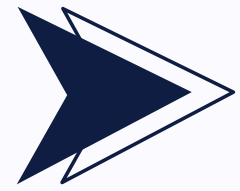
- February shows peak weather delay (1.0 min)
- Lowest delays occur in October (0.2 min)
- Delays rise again during winter months

Suggestions

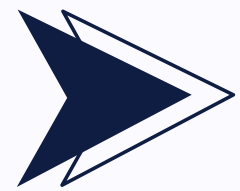
- Strengthen winter weather planning & monitoring
- Add buffer time and contingency flights in peak months



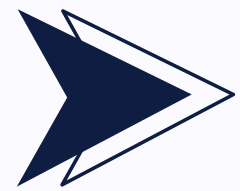
Tech Stack



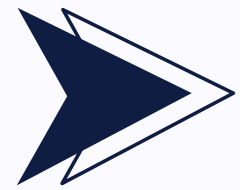
Data Handling: pandas, numpy



Visualization: matplotlib, seaborn, plotly, folium



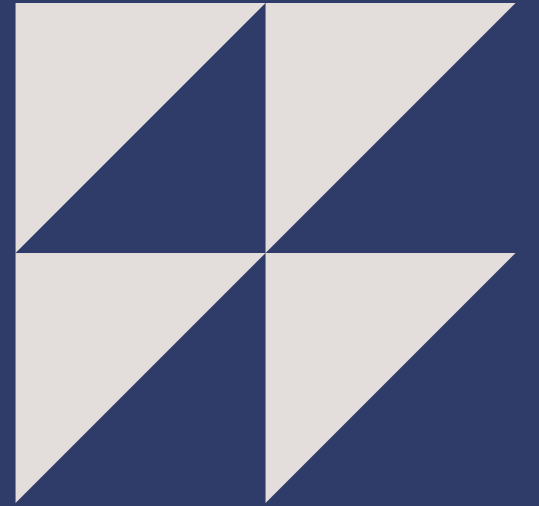
Optional Dashboard: Streamlit, PowerB, Markdown



Documentation: Jupyter Notebook, PDF, GitHub

Strategic Recommendations

Actionable steps to enhance data-driven decision-making and future growth



01 Maximize the "Morning Efficiency" Window

Prioritise high-value or time-sensitive flights for the 0-6 AM departure block, as this period consistently experiences the lowest average delays across all categories.

02 Implement Peak-Hour Buffers

Increase scheduling buffers and ground staff during late afternoon and evening blocks to mitigate the cumulative "cascade" effect, where initial disruptions multiply as the day progresses.

03 Proactive Winter Risk Management

Deploy additional de-icing resources and staff specifically for January and February, as cancellation rates nearly triple during these winter months (2.94%) compared to non-winter periods (1.11%)

Strategic Recommendations

Actionable steps to enhance data-driven decision-making and future growth



04 Targeted Hub Interventions

Focus operational review teams on "Hotspot" hubs like Chicago O'Hare (ORD) and Dallas/Fort Worth (DFW), which exhibit higher average delays across airline and air system categories.

05 Carrier-Specific Operational Audits

Since "Airline/Carrier" issues are a top reason for both delays and cancellations, major carriers should perform internal audits on crew scheduling and mechanical readiness to reduce controllable disruptions.

06 Seasonal Efficiency Opportunity

Leverage the September to December window for maximizing flight frequency and reliability, as these months show the most stable flying conditions and lowest cancellation rates.

Conclusion

This analysis highlights how data-driven insights can help airlines reduce delays, improve operational efficiency, and enhance customer satisfaction. By leveraging predictive analytics and proactive planning, airlines can achieve more reliable and profitable operations.

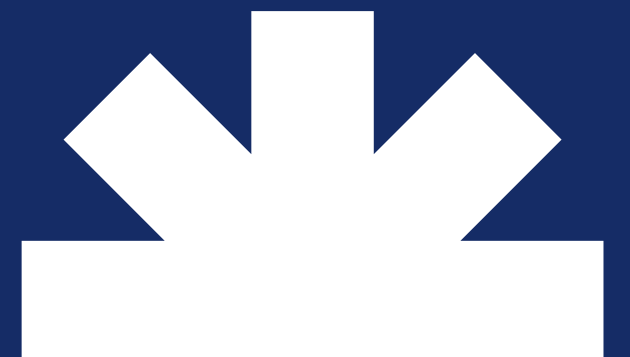
Website

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Any Questions?



Thank
you